

Single core Copper & Aluminum Conductors

XLPE Insulated, Aluminum Tape armored and PVC Sheathed

Voltage Grade

8.7 / 15 (17.5) KV

Description

Soft annealed Stranded circular compacted Copper or Stranded circular compacted Aluminum conductor, semiconducting layer as conductor screen, XLPE insulated semiconducting layer as non metallic insulation screen, Copper tape or wire as metallic insulation screen, armored with aluminum tape and PVC sheathed.

Application

These cables are generally suitable for direct burial or for installation on trays or in ducts.

Technical Data

Relevant Standard	IEC 60502
Conductor	Plain copper conductor class 2 or Aluminum conductor class 2
Conductor Screen	Extruded Semiconductive Compound
Insulation	Extruded XLPE Compound Rated 90° C
Insulation Screen	Non Metallic part: Extruded Semiconductive Compound (Bonded or Strippable) to the insulation Metallic part: (by one of the following) • Copper tape is applied helically with a suitable overlap. • Copper Wires helically applied and banded with a Copper tape to achieve electrical contact • Overlapped Copper Tape then Copper Wires are also available (in single core only) Conductor screen, XLPE insulation and non metallic insulation screen are applied at the same time using triple head extruder.
Inner Sheath	Extruded PVC Compound
Armoring	Double Aluminum Tapes applied helically over the inner sheath.
Outer Sheath	Extruded PVC Compound
Outer Sheath Color	Black (default color), or any other color according to customer request
Maximum Operating Temp.	90° C
Minimum bending Radius	12 x outer diameter
Packing Condition	Our standard length on Wooden Drum / or according to customer request



Product Code	Nominal Cross Sectional Area	Max. Conductor Resistance			Operating Capacitance	Inductance		Current Rating				Approx. Overall Diameter	Approx. Weight
		DC at 20 °C	AC at 90° C			Flat	Trefoil	Laid in Ground		Laid in Free Air			
			Flat	Trefoil				Flat	Trefoil	Flat Touched	Trefoil		
	mm²	Ω/km	Ω/km	Ω/km	μF/km	mH/km	mH/km	A	A	A	A	mm	kg/km
A - Copper Conductors													
PM-CT14-X7-01-DP	50	0.3870	0.4937	0.4937	0.2141	0.4936	0.4474	224	218	241	236	28.4	1245
PM-CT15-X7-01-DP	70	0.2680	0.3420	0.3421	0.2448	0.4664	0.4201	276	267	302	295	30.2	1490
PM-CT16-X7-01-DP	95	0.1930	0.2466	0.2466	0.2702	0.4466	0.4003	328	316	366	358	31.7	1765
PM-CT17-X7-01-DP	120	0.1530	0.1957	0.1958	0.2938	0.4292	0.3830	372	361	420	412	33.1	2025
PM-CT18-X7-01-DP	150	0.1240	0.1589	0.1590	0.3191	0.4169	0.3707	410	402	474	469	34.8	2420
PM-CT19-X7-01-DP	185	0.0991	0.1274	0.1276	0.3492	0.4042	0.3580	461	454	545	540	36.8	2825
PM-CT20-X7-01-DP	240	0.0754	0.0975	0.0978	0.3877	0.3881	0.3419	530	528	642	636	39.1	3400
PM-CT21-X7-01-DP	300	0.0601	0.0783	0.0787	0.4245	0.3783	0.3320	588	588	733	728	41.7	4070
PM-CT22-X7-01-DP	400	0.0470	0.0622	0.0628	0.4695	0.3670	0.3207	655	662	835	842	44.6	5035
PM-CT23-X7-01-DP	500	0.0366	0.0497	0.0505	0.5244	0.3555	0.3093	720	737	964	969	48.1	6165
PM-CT24-X7-01-DP	630	0.0283	0.0399	0.0409	0.5925	0.3471	0.3009	804	825	1073	1086	52.4	7560
PM-CT25-X7-01-DP	800	0.0221	0.0329	0.0342	0.6656	0.3374	0.2911	879	908	1210	1225	57.2	9475
B - Aluminum Conductors													
PM-AT14-X7-01-DP	50	0.6410	0.8219	0.8219	0.2209	0.4867	0.4405	171	170	185	184	28.8	975
PM-AT15-X7-01-DP	70	0.4430	0.5682	0.5682	0.2465	0.4640	0.4178	210	208	232	232	30.3	1085
PM-AT16-X7-01-DP	95	0.3200	0.4106	0.4106	0.2736	0.4443	0.3981	251	248	281	278	31.9	1210
PM-AT17-X7-01-DP	120	0.2530	0.3248	0.3248	0.2955	0.4405	0.3933	285	282	324	324	33.8	1360
PM-AT18-X7-01-DP	150	0.2060	0.2646	0.2646	0.3241	0.4145	0.3682	317	315	366	366	35.1	1550
PM-AT19-X7-01-DP	185	0.1640	0.2109	0.2111	0.3543	0.4021	0.3559	357	358	422	426	37.1	1730
PM-AT20-X7-01-DP	240	0.1250	0.1611	0.1613	0.3877	0.3881	0.3419	417	414	506	498	39.1	1945
PM-AT21-X7-01-DP	300	0.1000	0.1293	0.1295	0.4245	0.3783	0.3320	459	463	581	574	41.7	2220
PM-AT22-X7-01-DP	400	0.0778	0.1011	0.1015	0.4695	0.3670	0.3207	517	527	672	670	44.6	2670
PM-AT23-X7-01-DP	500	0.0605	0.0795	0.0800	0.5244	0.3555	0.3093	595	602	782	781	48.1	3110
PM-AT24-X7-01-DP	630	0.0469	0.0626	0.0633	0.5942	0.3468	0.3006	670	679	894	895	52.5	3695
PM-AT25-X7-01-DP	800	0.0367	0.0503	0.0512	0.6656	0.3374	0.2911	749	761	1025	1025	57.2	4405

The above data is approximate and subject to manufacturing tolerances

Note:-

Flame retardant PVC sheathed or Polyethylene Sheathed are available according to customer request
Longitudinal or radial Water tightness layers are also available according to customer request